

# COMPSCI 689

## Lecture 5: Generalization

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# Empirical Risk Minimization

Let  $\Theta$  be the parameter space for a set  $\mathcal{F}$  of prediction functions  $f_\theta$  mapping from  $\mathcal{X}$  to  $\mathcal{Y}$ . The principle of Empirical Risk Minimization (ERM) states that we should select the parameters  $\theta \in \Theta$  that *minimize the average of the prediction loss*  $L(\mathbf{y}_n, f_\theta(\mathbf{x}_n))$  computed over the data set  $\mathcal{D}$ , also known as the empirical risk  $R(f_\theta, \mathcal{D})$ :

$$\hat{\theta} = \arg \min_{\theta \in \Theta} R(f_\theta, \mathcal{D})$$

$$R(f_\theta, \mathcal{D}) = \frac{1}{N} \sum_{n=1}^N L(\mathbf{y}_n, f_\theta(\mathbf{x}_n))$$

**Question:** How well should we expect prediction functions learned using ERM to perform on new data?

# Data Generating Distributions

- We introduced the concept of a data set  $\mathcal{D}$ , but we have not made any assumptions about the process that generates the data set.
- The most common assumption is that data cases are generated independently and identically (IID) from an underlying joint probability distribution over  $(\mathbf{x}, y)$  pairs:  $p_*(\mathbf{X} = \mathbf{x}, Y = y)$ .
- We refer to  $p_*$  as the *true data generating distribution* or *true generative process*.

# Data Generating Distributions

- Using the chain rule of probability, we can write:

$$p_*(\mathbf{X}_n = \mathbf{x}_n, Y_n = y_n) = p_*(\mathbf{X}_n = \mathbf{x}_n)p_*(Y_n = y_n|\mathbf{X}_n = \mathbf{x}_n)$$

- Under the IID assumption, the probability distribution of a data set  $\mathcal{D}$  sampled from  $p_*$  is given by:

$$\begin{aligned} p_*(\mathcal{D}) &= \prod_{n=1}^N p_*(\mathbf{X}_n = \mathbf{x}_n, Y_n = y_n) \\ &= \prod_{n=1}^N p_*(\mathbf{X}_n = \mathbf{x}_n)p_*(Y_n = y_n|\mathbf{X}_n = \mathbf{x}_n) \\ &= p_*(\mathbf{X})p_*(\mathbf{Y}|\mathbf{X}) \end{aligned}$$

- ... where  $p_*(\mathbf{X}) = \prod_{n=1}^N p_*(\mathbf{X}_n = \mathbf{x}_n)$
- ... and  $p_*(\mathbf{Y}|\mathbf{X}) = \prod_{n=1}^N p_*(Y_n = y_n|\mathbf{X}_n = \mathbf{x}_n)$

# Example: Data Generating Distributions

- Consider the regression setting where  $x \in \mathbb{R}$  and  $y \in \mathbb{R}$ .
- Let  $p_*(X = x) = U(x; 0, 1)$ : the uniform probability density on  $[0, 1]$ .
- Let  $p_*(Y = y|X = x) = N(y; xw + b, \sigma^2)$ : a conditional normal density on  $y$  given  $x$ .
- Then the joint density on  $x$  and  $y$  is given by:

$$p_*(X = x, Y = y) = U(x; 0, 1) \cdot N(y; xw + b, \sigma^2)$$

- To produce a dataset  $\mathcal{D}$  of  $N$  samples  $(x_n, y_n)$ , for each  $n$  we can sample  $x_n \sim U(x; 0, 1)$  and then sample  $y_n \sim N(y; x_n w + b, \sigma^2)$ .
- The probability of the data set is:

$$p_*(\mathcal{D}) = \prod_{n=1}^N U(x_n; 0, 1) \cdot \mathcal{N}(y; x_n w + b, \sigma^2)$$

# Expected Prediction Loss Minimization

- If we had direct access to  $p_*$ , we could attempt to directly minimize the expected prediction loss instead of the empirical risk:

$$\hat{\theta} = \arg \min_{\theta} E_{p_*(y, \mathbf{x})} [L(y, f_{\theta}(\mathbf{x}))]$$

$$E_{p_*(y, \mathbf{x})} [L(y, f_{\theta}(\mathbf{x}))] = \int_{\mathcal{X}} \int_{\mathcal{Y}} p_*(y, \mathbf{x}) L(y, f_{\theta}(\mathbf{x})) d\mathbf{x} dy$$

$$E_{p_*(y, \mathbf{x})} [L(y, f_{\theta}(\mathbf{x}))] = \int_{\mathcal{X}} \sum_{y \in \mathcal{Y}} p_*(y, \mathbf{x}) L(y, f_{\theta}(\mathbf{x})) d\mathbf{x}$$

- However, in a standard learning setting, our only information about  $p_*$  is provided through the  $(\mathbf{x}_n, y_n)$  *samples* from  $p_*$  that comprise the data set  $\mathcal{D}$ , leading us to select the optimal prediction function  $f_{\hat{\theta}}$  using ERM.

# Future Data

- While a prediction function  $f_{\hat{\theta}}$  found using ERM is optimal on the data set used to learn the model, what we really care about is expected performance with respect to the distribution of future data  $p'_*$ .
- For a prediction function  $f_{\hat{\theta}}$  learned on samples from  $p_*$  to *generalize* to future data sampled from a distribution  $p'_*$ , the distributions  $p_*$  and  $p'_*$  must be related in some way.
- The most common assumption is simply that  $p'_* = p_*$ .
- When  $p'_* = p_*$ , the future data are referred to as being *in-distribution*.

# Future Data

- When  $p'_* \neq p_*$ , the future data are referred to as *out of distribution* or OOD.
- Detection and/or robustness to OOD instances is a critical issue in many current machine learning applications where systems are deployed “in the wild.”
- In many cases,  $p'_*$  may also not be static over time. In particular,  $p'_*$  may drift away from  $p_*$  over time resulting in the need for *continual learning*.



# Expected Prediction Loss

- **Question:** How can we estimate generalization performance?
- If we had access to the true generating distribution of the future data, the statistic we are interested in is the expected value of the prediction loss of our learned model under this distribution.
- Note that when considering generalization, we can do so under a prediction loss function  $L'$  that differs from the prediction loss used to train the model.

# Expected Prediction Loss

- Under the assumption that the future data are generated by  $p'_*$ , the expected prediction loss of a model with estimated parameters  $\hat{\theta}$  is given by:

$$E_{p'_*(y, \mathbf{x})}[L(y, f_{\hat{\theta}}(\mathbf{x}))] = \int_{\mathcal{X}} \int_{\mathcal{Y}} p'_*(y, \mathbf{x}) L'(y, f_{\hat{\theta}}(\mathbf{x})) d\mathbf{x} dy$$

$$E_{p'_*(y, \mathbf{x})}[L(y, f_{\hat{\theta}}(\mathbf{x}))] = \int_{\mathcal{X}} \sum_{y \in \mathcal{Y}} p'_*(y, \mathbf{x}) L'(y, f_{\hat{\theta}}(\mathbf{x})) d\mathbf{x}$$

- However, as when learning models, we typically don't have access to the distribution of future data  $p'_*$ .

# Train/Test Experiment

- To learn a model and obtain an unbiased estimate of generalization performance, we can use a Train/Test experiment.
- We use a data set  $\mathcal{D}_{tr}$  sampled from  $p_*$  referred to as the *training set* to learn the model parameters  $\hat{\theta}$  using ERM with the prediction loss  $L$ . We then evaluate performance of  $f_{\hat{\theta}}$  using  $L'$ .
- We use a second data set sampled from  $p'_*$  to estimate the performance of the learned model on future data with respect to the prediction loss  $L'$ . This second data set is referred to as the *test set* and is denoted  $\mathcal{D}_{te}$ .
- If we only have a single data set  $\mathcal{D}$ , we can split it into a training set  $\mathcal{D}_{Tr}$  of size  $N_{tr}$  and a test set  $\mathcal{D}_{Te}$  of size  $N_{te}$  at random.
- However, this is equivalent to assuming that  $p_* = p'_*$ , which may or may not be true.

# Train/Test Performance Evaluation

Given a learned model  $f_{\hat{\theta}}$  trained using a prediction loss function  $L$ , we can evaluate the model using any test prediction loss function  $L'$  that we choose, including functions  $L'$  that are discontinuous and non-differentiable.

■ Average Test Loss:  $\frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{Te}} L'(y, f_{\hat{\theta}}(\mathbf{x}))$

# Common Test Losses

- Mean Squared Error (MSE):  $\frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{te}} (y - f_{\hat{\theta}}(\mathbf{x}))^2$
- Mean Absolute Error (MSE):  $\frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{te}} |y - f_{\hat{\theta}}(\mathbf{x})|$
- Error Rate:  $\frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{te}} [y \neq f_{\hat{\theta}}(\mathbf{x})]$

# Properties of Average Test Loss

- Under the assumption that  $\mathcal{D}_{te}$  is sampled IID from  $p'_*$  the average test loss computed using a prediction loss function  $L'$  is an unbiased estimate of the expected prediction loss using  $L'$ .

$$\mathbb{E}_{p'_*(\mathcal{D}_{te})} \left[ \frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{te}} L'(y, f_{\hat{\theta}}(\mathbf{x})) \right] = E_{p'_*(y, \mathbf{x})} [L'(y, f_{\hat{\theta}}(\mathbf{x}))]$$

- Importantly, using the same data set to both learn the model parameters  $\theta$  and compute the test loss will lead to estimated performance that is biased low.
- In other words, testing on the training set will overestimate how well the learned model generalizes.
- Bottom line: **Never test on train.**

# Properties of Average Test Loss

- Next, under the assumption that  $\mathcal{D}_{te}$  is sampled IID from  $p'_*$ , by the law of large numbers, the average test loss will converge to the expected prediction loss as  $N_{te}$  goes to infinity.

$$\lim_{N_{te} \rightarrow \infty} \frac{1}{N_{te}} \sum_{(\mathbf{x}, y) \in \mathcal{D}_{Te}} L'(y, f_{\hat{\theta}}(\mathbf{x})) = E_{p'_*(y, \mathbf{x})}[L'(y, f_{\hat{\theta}}(\mathbf{x}))]$$

- **Question:** What can we say in the finite sample case?

# Finite Sample Linear Regression Expected Error

To analyze the closed-form solution to linear regression, we will make the following assumptions. This setting is often referred to as the “well-specified model” setting.

- $p_* = p'_*$
- $p_*(y|\mathbf{x}) = \mathcal{N}(y; \mathbf{x}\theta_*, \sigma_*^2)$
- $p_*(\mathbf{x}) = \mathcal{N}(\mathbf{x}; 0, \Sigma_*)$
- We take  $L'$  to be the squared prediction loss.

Under these assumptions, it is easy to see that for any  $\mathbf{x}$ , the expected squared loss given the true parameters  $\theta_*$  is:

$$\mathbb{E}_{p_*(y|\mathbf{x})}[(y - \mathbf{x}\theta_*)^2] = \sigma_*^2$$



# Finite Sample Linear Regression Expected Error

Assuming that we fit the linear regression model parameters  $\hat{\theta}$  using squared loss with a data set  $\mathcal{D}_{tr}$  of size  $N_{tr} \gg 1$  and the dimension of  $\mathbf{x}$  is  $D$ , we have the following result:

$$\mathbb{E}_{p_*}[(y - \mathbf{x}\hat{\theta})^2] = \sigma_*^2 \left( 1 + \frac{D}{N_{tr} - D - 1} \right)$$

For example, if  $D = 20$  and  $N = 100$ , we have that the expected prediction loss of  $f_{\hat{\theta}}$  is about  $1.25\sigma_*^2$ .

In general, we can see that with  $D$  fixed, as  $N_{tr}$  goes to infinity, the expected loss of  $f_{\hat{\theta}}$  converges to  $\sigma_*^2$ .

# Finite Sample Classification Error Bounds

- Consider the classification case taking  $L'$  to be the zero-one loss.
- Let  $\pi_* = E_{p'_*(y, \mathbf{x})}[y \neq f_{\hat{\theta}}(\mathbf{x})]$  be the true expected classification error rate.
- Suppose we observe an empirical classification error rate of  $\pi_e$  on a test data set  $\mathcal{D}_{te}$  of size  $N_{te}$ , then for any  $0 < \delta \leq 1$  we have:<sup>1</sup>

$$p'_* \left( \pi_* \leq \pi_e + \sqrt{\frac{\log(1/\delta)}{2N_{te}}} \right) \geq 1 - \delta$$

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<sup>1</sup>Langford. Tutorial on Practical Prediction Theory for Classification. JMLR 2005.

## Example: Classification Error Bounds

Suppose  $N_{te} = 1000$  and  $\delta = 0.01$

$$p'_* \left( \pi_* \leq \pi_e + \sqrt{\frac{\log(1/\delta)}{2N_{te}}} \right) \geq 1 - \delta$$

$$p'_* \left( \pi_* \leq \pi_e + \sqrt{\frac{\log(1/0.01)}{2 \cdot 1000}} \right) \geq 1 - 0.01$$

$$p'_* \left( \pi_* \leq \pi_e + \sqrt{\frac{\log(100)}{2000}} \right) \geq 0.99$$

$$p'_* (\pi_* \leq \pi_e + 0.047985) \geq 0.99$$